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Indian J Psychiatry. 2026 Jan 27;68(1):68–93. doi: [10.4103/indianjpsychiatry_1321_25](https://doi.org/10.4103/indianjpsychiatry_1321_25)

Clinical practice guideline for assessment and management of depression in India

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PMCID: PMC12900049 PMID: [41694045](#)

Abstract

This Clinical Practice Guideline has aimed to provide mental health practitioners a comprehensive and evidence-based framework that can be used for the assessment and management of depression specifically catered to the Indian population. Considering the high prevalence of depression and its burden on the affected persons as well as the medical system, these guidelines were needed to help provide psychiatric care that matches the international standards and the cultural and practical needs of the Indian population. The article outlines key risk factors, components of assessment, and pharmacological and psychological management, along with structured approaches for suicide risk management. It also seeks to outline recommended investigations, differential diagnosis, and clinical decision guidelines based on the severity of the psychopathology.

Guidelines are offered for the selection of antidepressants, treatment phases, monitoring, switching or augmenting strategies, and tapering the pharmacological treatment. Management strategies for comorbidities, special populations, and treatment-resistant depression are offered in this article. Newer treatment modalities including neuromodulation and novel pharmacological agents showing promise are discussed. Overall, the guideline aims to support clinicians in delivering personalized, effective, and culturally appropriate care for individuals with depression in India.

Keywords: Clinical practice guideline, depression, treatment and management

INTRODUCTION

Depression is one of the most common psychiatric disorders. It is also one of the most debilitating disorders owing to high prevalence and significant impact on disability-adjusted life-years (DALY).^[1] Despite it being a well-recognized disorder, problems exist in its identification, diagnosis, treatment, and long-term management. The prevalence of depression in India ranges from 3% to 43% according to various studies,^[2] with the WHO ranking it as the foremost contributor to the global burden of disease in the near future.^[3] The National Mental Health Survey (NMHS) of India highlighted a substantial treatment gap for depression.^[4] According to analyses of the 2015–16 survey data, about 79.1% of individuals with current depressive disorders had not received care from a formal healthcare provider.^[5]

The field of psychiatry is taking the high prevalence rate of mental disorders in account and is changing and developing rapidly. There is an advent of new pharmacological agents, neuro-modulation strategies, and advanced psychotherapeutic interventions that can cater to the growing need for mental health care among the Indian masses. The dynamic nature of the field necessitates timely and culturally adaptable Clinical Practice Guidelines that will aid mental health practitioners in delivering evidence-based care to their patients. Additionally, given the heterogeneity in the way clients present with their symptoms, comorbidities, and socio-occupational impairments, flexible yet standardized guidelines become paramount to ensure adequate tailored treatment for people from various demographics.

Institutions like the American Psychological Association (APA), National Institute for Health and Care Excellence (NICE), and Canadian Network for Mood and Anxiety Treatments (CANMAT) publish clinical practice guidelines, incorporating both the pharmacological and nonpharmacological treatment modalities that emphasize not only personalization of care but also shared decision-making approaches, ultimately promoting interdisciplinary research and collaboration between different professionals. In India, the Indian Psychiatric Society (IPS) publishes its own clinical practice guidelines (CPG) that are informed by the cultural and logistical considerations that include but are not limited to psychosocial contexts, somatic symptomatology, comorbid medical conditions, and caregiver burden. Nevertheless, a significant treatment gap persists because of a lack of resources, access to mental health services, limited continuing medical education (CME) for practicing physicians/psychiatrists, stigma and discrimination, the presence of traditional healers and faith healers, and so on.

The aim of the current CPG is to update existing guidelines based on international standards, but standardized as per the national needs to advance psychiatric care available to the Indian population. Special attention is given to diagnostic clarity, first- and second-line treatment strategies, chronic and treatment-resistant depression, and the advancements in the available treatment modalities. The overarching goal of the guidelines is to bridge the gap between global and local psychiatric practices, enabling professionals to provide personalized care for their patients with depression. Moreover, it is intended to provide a comprehensive framework for assessing and managing cases. Clinical situations often require individualized decision-making based on patient-specific factors that may not be fully captured in any guideline. Therefore, these recommendations are to be viewed as broad guidelines rather than legally binding, and clinicians are encouraged to exercise their professional judgment in tailoring care to the unique needs of each patient. Treatment may be adapted by clinicians according to available resources and the context of the treatment setting. Additionally, these guidelines are intended primarily for patients diagnosed with depressive disorders according to ICD-11 or DSM-5 criteria. Management of other conditions, including subthreshold depressive symptoms, dysthymia, post-psychotic depression, or depressive symptoms occurring in other clinical contexts, may differ and should be based on individual clinical assessment and professional judgment.

RISK FACTORS

The development of depression is shaped by both nonmodifiable and modifiable risk factors. Nonmodifiable factors include female gender, family history of mood disorders, adverse childhood experiences, and loss of a spouse. Modifiable factors include chronic medical illnesses, psychiatric comorbidities, alcohol or substance use, insomnia, hormonal changes [e.g., premenstrual dysphoric disorder (PMDD), hypothyroidism, peripartum depression], stressful life events, job strain, bullying, sedentary lifestyle, and excessive screen time. Awareness of these determinants helps in targeted screening, especially in high-risk populations.[\[6\]](#)

PRINCIPLES OF ASSESSMENT AND DIAGNOSIS

Given the high heterogeneity in the manifestation of depressive symptoms, it becomes pertinent to cater to the individual presentation of the same underlying condition. Symptoms may be masked or subthreshold, leading to functional impairment despite not meeting the full diagnostic criteria. The aim of the initial assessment is to establish a diagnosis and negotiate a plan for further assessments, management, and monitoring. A comprehensive diagnostic assessment should integrate biological, psychological, and social factors, while being culturally sensitive and longitudinal, and should incorporate family inputs whenever appropriate [\[Figure 1\]](#).

Figure 1.

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Assessment and diagnostic approach for depression. *BMI= Body Mass Index, CBC=Complete Blood Count, TSH= Thyroid-Stimulating Hormone, LFT=Liver Function Test, RFT=Renal Function Test, PHQ= Patient Health Questionnaire, HAM-D= Hamilton Depression Rating Scale, MADRS= Montgomery-Åsberg Depression Rating Scale

Core components of assessment

- Detailed history, physical examination, and mental status examination (MSE)
- Exploration of emotional, cognitive, behavioral, and somatic symptoms
- Evaluation of risk and resilience factors across the lifespan
- Collateral information from family/caregivers

A myriad of tools are available to assess the symptom severity in addition to the clinical interview, which include but are not limited to: Beck Depression Inventory (BDI),^[7] Hamilton-Depression Rating Scale (HAM-D),^[8] The Depression, Anxiety, and Stress Scale (DASS),^[9] Patient Health Questionnaire-9 (PHQ-9),^[10] and Montgomery-Asberg Depression Rating Scale (MADRS).^[11] Similarly, different scales are available for different populations (Geriatric Depression Scale (GDS),^[12] Children's Depression Inventory (CDI),^[13] Children's Depression Rating Scale (CDRS),^[13] and so on).

Additionally, several indications of depressive symptoms can be found in the MSE itself, which is conducted on each patient at the time of intake. These indications may be manifested in the form of a poorly groomed general appearance (e.g., downcast gaze, furrows between eyes, omega sign, veraguth' fold); agitated or retarded motor behavior; soft, slow, or monotonous speech with delayed reaction time, and scant productivity; depressed or irritable affect (sometimes anxious mood might also prevail), constricted, reduced reactivity; and retarded thought stream and thought content of hopelessness, worthlessness, helplessness, ideas of guilt, and suicidal ideas. Cognitively, attention, concentration, and memory might be impaired. Judgement may also be impaired in certain conditions with partial to full insight. The presence of psychotic symptoms may change the MSE presentation of the patient. A careful evaluation is helpful in differentiating psychotic depression from other psychotic conditions.

Suicidality

A thorough suicide assessment is indicated for each patient with depression. It has been found that the prevalence of self-injury or suicidal behavior may be as high as 52% in adolescents with depression.^[14] These rates may vary as per the socio-demographic factors, but they are alarmingly high. Beck's Suicide Intent Scale^[15] and Columbia-Suicide Severity Rating Scale (C-SSRS)^[16] are commonly used scales that can be applied to patients coming in for the treatment of depression. There are specific treatment needs

of patients presenting with suicidality. Patients with high risk can be better managed in an inpatient setting.

Suicide-risk assessment in depression

A systematic review and meta-regression study found that the predicted adjusted relative risk of suicide is 4.11 (2.1–8.1) and 7.64 [4.3, 13.6] in dysthymia and major depressive disorders, respectively.[\[17\]](#) Suicide risk assessment in a depressed patient, like suicide risk assessment in general, involves a collaborative and comprehensive evaluation, keeping the bio-psycho-social model of depression and suicide risk in mind. The clinical assessment should explore the risk factors, protective factors, current psychiatric diagnoses, past psychiatric and medical history, and personal history, including psychosocial stressors and family history (refer to [Table 1](#)).

Table 1.

General considerations while dealing with an individual with suicidal behavior

- Assessment of suicidal behavior in depression involves a thorough enquiry in severity of depression, suicide risks, and comorbid psychiatric illness.
- The assessment involves a collaborative therapeutic alliance, and a shared understanding of the patients’ suicidal state.
- Establishing the diagnosis is vital in guiding the short-term and long-term treatment.
- Ensuring patients’ immediate safety and accordingly selecting treatment and its setting.
- Safeguarding patients’ health and wellbeing must be the goal of the treatment.
- Family members and significant others be involved in assessment, whenever indicated
- Information should be obtained from all possible sources (family members, medical records, previous physicians involved, social networks, etc.)

- Not all information is immediately needed; the focus should be on the information required in acute management
 - Obtaining patients' consent while involving others in the assessment
 - Ensuring confidentiality and conveying the same to the patients explicitly
 - The confidentiality can be breached if there is an imminent danger to the patients
 - Assess patients' vulnerabilities and strengths (past, current, and future histories, including risk and protective factors) and tailor interventions accordingly
 - Suicide rating scales can add to diagnoses and risk estimation but cannot be solely relied upon.
 - Setting of treatment should be in the least restrictive environment (MHCA, 2017).[18]
 - The treatment under section 89, MHCA should be as per the advanced directive (AD) of the patients in collaboration with the nominated representatives (NR). However, if the AD is not available, the attending clinician can make a treatment decision in collaboration with the caregivers of the patient. Further, during an emergency, the AD does not apply.
 - The intensity and setting of the treatment should also be governed by comorbid mental and physical illnesses, availability, and strength of the psychosocial networks.
 - The patients' ability to maintain adequate self-care, provide reliable feedback, and cooperate with the treatment is a critical factor in determining the treatment setting.
 - Documenting suicide assessment is vital, including conversation with patients, and caregivers.
 - Details of the suicidal behaviors, prospective risk and protective factors, and advises and intervention offered, including any refusal to treatment, must also be documented.
 - The MHP providing care to individuals with recurrent suicidal behavior needs to be aware of their own reaction to the patients' condition that may affect the treatment.
 - The MHP should address distress, self-doubt, and resource needs among clients through peer support, supervision, and feedback.
-

*F/m - family members, MHCA - Mental Healthcare Act, 2017

There are various risk factors for suicidal behavior and completed suicide in depression that need to be assessed during suicide risk assessment. Literature suggests that the risk factors for suicidal ideation are severity of hopelessness and depression with sleep disturbances (e.g., nightmares, insomnia).[\[19,20\]](#)

A history of suicide attempts, suicidal ideation, severe depression, psychotic symptoms, the severity of hopelessness predicts suicide attempt, subsequent suicide attempts, time to full remission of depression, and sleep issues are relevant predictors of future suicide attempts. Other risk factors are mental pain, poor interpersonal and communication skills, alexithymia, lack of social support, impaired cognitive control, poor decision-making, and impulsivity-aggression traits.[\[21\]](#)

The risk factors for completed suicide in individuals with depression are male gender, family history of psychiatric disorder, previous attempted suicide, and severe depression, including with psychotic symptoms, hopelessness, associated sleep issues, and comorbid disorders, including anxiety and misuse of alcohol and drugs.[\[19,22\]](#)

Additionally, patients' socio-economic conditions and cultural and religious beliefs need to be accounted for in suicide risk assessment [\[Table 2\]](#). A thorough risk assessment would inform the setting, choice, and intensity of treatment and psychosocial interventions. In [Table 1](#), we have provided general considerations during suicide-risk assessment.

Table 2.

Psychiatric evaluation of an individual with depression and suicidal behavior

Assessment of Current Suicidality

- Suicidal ideations (intensity, frequency, controllability, content, timing, triggering and relieving factors)
- Suicidal plan (elaborate plans present or not? intent to act on them)
- Suicidal behavior (timing, intent, consequences of the behavior, whether aborted or interrupted attempts, any preparatory acts, circumstances)
- Suicidal intent
- Methods considered for suicide (lethality, patients' expectation about the lethality, enquiring about the availability of lethal means, e.g., firearms, pesticides)
- Assess patients' knowledge and skill in using the means
- Evidence of state or trait markers of hopelessness*, impulsivity, anhedonia, agitation
- Reasons for being alive and future
- Any thoughts or plans or intentions to harm others
- Interpersonal aspects or dynamics of suicidal behavior should be delineated
- Patient's thoughts about the suicidal attempt (remorse, level of premeditation, reason for choosing a means, current understanding about life and death)
- Perceived ability to act on suicidal ideations

Assess for Comorbid Psychiatric Disorders

- Apart from depression, symptoms and signs of other mental illnesses like schizophrenia and psychotic illness, SUDs, anxiety disorders (e.g., panic disorder), personality disorders (e.g., borderline personality disorder) need to be explored
- Previous psychiatric history (onset, severity, course), including for substance use disorders, and related hospitalization

Treatment History and Treatment Relationships

- Previous therapeutic relationship with the mental health professionals or physicians
- Gauge current level of the therapeutic alliance with the treatment

provider(s)

- Impact of these relations with the current suicidal behavior and future care

Past and Family History

- Previous history of suicidal behavior (suicide attempt, including aborted or interrupted ones, frequency, recency, and severity) or self-harming behavior, including patients' involvement with the circumstances
- Previous or current physical illnesses and treatment, including surgery and hospitalization for the same
- Family history of suicide or suicide attempt, psychiatric disorders, including SUDs

Psychosocial Aspects

- This is a significant aspect of psychiatric evaluation for suicidality in depression, in the Indian context and has implications for the immediate management
- Acute psychological crises (relational issues, financial difficulty, loss of loved ones, changes in the socio-economic status)
- Chronic psychosocial stressors (interpersonal difficulties, financial difficulties, family discord, domestic violence, ongoing or history of physical or sexual abuse)
- Feeling of entrapment, thwarted belongingness, perceived burdensomeness; ruminations, self-criticism, ideas of guilt
- Psychodynamic issues are vital for a comprehensive assessment
- Employment status, current living arrangements, extent of external supports
- Religious and cultural beliefs or practices around life, death and suicide

Individual Strengths/Protective Factors and Vulnerabilities/Risk Factor

- Individuals' coping skills
- Personality traits (impulsivity/risk taking behavior, emotional dysregulation, constricted/polarized thinking, perfectionist tendencies are at risk for suicidality)
- Previous adjustment with or response to stress, including cognitive

appraisals

- Capacity for reality testing
 - Capacity to tolerate psychological pain (e.g., loneliness) and satisfy emotional needs
 - Exploring modifiable and irreversible risk factors are crucial
-

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*Hopelessness may range from negative expectations for the future to being devoid of hope or despairing for the future. Depression with hopelessness carries a greater suicide risk. SUD - substance use disorders

The detailed evaluation of the psychiatric condition of the patients and their suicidal behavior should involve their current suicidal ideations, planning, suicidal intention, methods used/being considered, lethality, patients' perception of that lethality, and patients' skill in using those means. In [Table 1](#), we have also provided a detailed account of the psychiatric evaluation and suicidal behavior in depressed patients with suicidality.

The risk assessment of suicide attempt in individuals with suicidal ideation needs deeper exploration. This includes severity of depression, current substance use, chronic pain, prominent anxiety or agitation, impulsivity, history of hospitalization in the previous year, suicide attempt in the previous year, recent history of mental health issue-related emergency visits, adverse childhood experiences, history of deliberate self-harm, family history of suicide, recent stressor, and current social support.[\[23,24\]](#)

SUICIDE RISK ASSESSMENT INSTRUMENTS

In clinical practice, a psychiatrist or mental health professional relies on patients' self-reports; however, for a comprehensive risk estimation account of the family members, social networks and treating teams involved in previous provision of care to the patients should be obtained. Various scales have been developed for the suicide risk estimation and also cross-culturally validated in depression (e.g., Columbia Suicide Severity Rating Scale,

Ask Suicide Severity Questions); their use needs to be strengthened by performing a comprehensive history taking and mental status examination[25] [Table 3].

Table 3.

Characteristics of Scales commonly used to measure suicidal behavior

Scale	Scale characteristics	Remark
Beck's Scale for Suicidal Ideation (BSSI)[26]	• 19-item scale • Enquires about the presence and intensity of suicidal thoughts in the last week • Contains five screening items (wish to live and die, reason for living/dying, desire to make an active suicide attempt, and passive suicide desire). • It has domains like suicidal thoughts, plan, and intent to commit suicide. Item score varies from 0 to 2. • Factor structure – three factors: active suicide desire, passive suicide desire, and preparation.	• Cross-culturally validated • Among the most widely used instrument to assess suicidal ideations • Concern over predictive validity
Suicidal intent scale (SIS)[27]	• 15-item scale • Evaluates factual aspects of the most recent suicide	• Considers both subjective and objective circumstances around the

attempt (the patient's thoughts, feelings and circumstances around the suicide attempt).

- Factor structure –
- Lethal Intent factor contained items pertaining to the subjective level or personal experience of lethal intent (item 9 to 15)
- Planning factor contained items largely related to objective planning or circumstances around the attempt (item 1-8).

event, hence provide a reliable picture of suicidal behavior

- Lack predictive validity

Columbia Suicide Severity Rating Scale (C-SSRS) [28,29]

- Most used and the standard for suicide risk assessment in mental health settings
- comprehensively assesses the suicidal ideation, intensity of suicidal ideation, suicidal behavior and lethality around it.
- Evaluates the presence, severity, and frequency of suicidal thoughts and behavior
- Distinguishing between suicidal and nonsuicidal

- Apart from Suicide-risk assessment, valuable in monitoring treatment outcomes, and guiding clinical management decisions.

self-injury

- Quantifies risk since the last visit to the healthcare facility.

Ask Suicide-Screening Questions (ASQ) [30,31]

- Brief suicide risk screening tool, having just four items
- Self-report instrument
- Available in Hindi
- Items include suicidal ideation, perceived burdensomeness on others and history of suicide attempt
- Participants who screened positive for suicide risk are then asked an 'acuity' question (current suicidal ideation)
- If an individual is also positive on this item, they are labeled as 'acute positive' screen for suicide; however, if negative for this item, they are labeled as 'nonacute positive' screen
- Has some predictive validity
- Has been found to be valid across the health setting, medical issues, and age groups

PHQ-9, 9th item [32,33]

- Evaluates passive thoughts of death or self-injury within the last 2 weeks
- PHQ-9, 9th item has predictive utility of suicide attempt and death by suicide on both immediate and long-term bases across age groups.

Some of the most used scales to measure suicidality in general and suicidal behavior in depression are Beck's scale for suicide ideation (BSSI), Suicidal intent scale (SIS), the Columbia suicide severity rating scale, Ask Suicide-Screening Questions (ASQ), and the Patient Health Questionnaire-9, 9th Item.

TREATMENT HISTORY

If the patient has received treatment in the past, it is essential to evaluate the information, including the type of antidepressant used, dose of psychotropic medication, compliance with medication, reasons for poor compliance, reasons for discontinuation of medication, response to treatment, and side effects experienced. If the medications were changed, then the reason for the change is also to be evaluated [[Table 4](#)].

Table 4.

Medications reported to induce or exacerbate depression [[34](#),[35](#),[36](#)]

Category	Examples
Cardiovascular drugs	Angiotensin-Converting Enzyme inhibitors, calcium channel blockers, clonidine, digoxin, hydralazine, methyl dopa, procainamide, propranolol, reserpine, thiazide diuretics, guanabenz
Chemotherapeutics	6-Azauridine, asparaginase, azathioprine, bleomycin, cisplatin, cyclophosphamide, doxorubicin, vinblastine, vincristine
Antiparkinsonian drugs	Amantadine, bromocriptine, levodopa
Stimulants (usually	Amphetamines, caffeine, cocaine, methylphenidate

during withdrawal)

Anti-microbial agents	Ampicillin, chloramphenicol, chloroquine, clofazimine, cycloserine, cyclosporine, dapsone, ethambutol, ethionamide, foscarnet, ganciclovir, griseofulvin, isoniazid, metronidazole, nalidixic acid, nitrofurantoin, penicillin G procaine, streptomycin, sulfonamides, tetracycline, trimethoprim
CNS depressants	Barbiturates, benzodiazepines, chloral hydrate, ethanol
Antiretrovirals	Efavirenz, enfuvirtide, saquinavir, zidovudine
Hormonal agents	Adrenocorticotropin, anabolic steroids, glucocorticoids, oral contraceptives
Antipsychotics	Fluphenazine, haloperidol
Other	Cimetidine, disulfiram, ethosuximide, phenobarbital, phenytoin, primidone, tiagabine, vigabatrin, tamoxifen, NSAIDs

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*CNS: Central Nervous System

Physical examination in assessment

Given the strong association between major depressive disorder (MDD) and various medical illnesses, a focused physical examination and basic investigations are essential to rule out nonpsychiatric causes of depression. Routine laboratory testing is not universally indicated; however, complete blood count (CBC) and thyroid-stimulating hormone (TSH) are low-cost tests that can help detect anemia and thyroid dysfunction, both of which are modifiable factors to low mood and fatigue. Additional investigations (e.g., testosterone, Vitamin D, Vitamin B12) should be ordered only when suggested by clinical history or examination findings or in individuals with relevant comorbidities, SUD, advanced age, self-neglect, or impaired judgment/communication.

Similarly, there is no evidence to support routine use of electrocardiography (ECG), electroencephalography (EEG), or neuroimaging in all patients with MDD. These should be performed only when there is a specific clinical indication:

- **ECG** – in patients with cardiovascular disease or when prescribing QTc-prolonging medications (*Citalopram, Most Tricyclic Antidepressants, adjuvant Thioridazine, Ziprasidone, Quetiapine, Chlorpromazine, etc.*).
- **Neuroimaging** – in cases with neurological signs, new or persistent cognitive impairment, or sudden changes in mood, behavior, or personality; also, in new-onset late-life depression to exclude cerebral events or structural brain lesions (e.g., tumor, metastasis).

Comorbidity

See [Table 5](#) for details of (psychiatric as well as physical illnesses) comorbidities commonly observed with depression.

Table 5.

Prevalence of comorbidities in MDD

Comorbidity	Prevalence Rate	Comorbidity	Prevalence Rate
Anxiety	67% [37]	Autoimmune Diseases	50% [38]
IBS	37% [39]	OCD	40-80% [40]
SUD	25% [41]	Eating disorders	93% [42]
ADHD	53% [43]	Epilepsy	29.3% [44]
Parkinson's Disease	40-50% [45]	Multiple Sclerosis	25.3% [44]
Metabolic Syndrome	25%-44% [46]	Alzheimer's Disease	90% [47]
Hypertension	27% [48]	Coronary Artery Disease	20.8% [49]

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ADHD: Attention Deficit Hyperactivity Disorder; IBS: Irritable Bowel Syndrome; MDD: Major Depressive Disorder; OCD: Obsessive Compulsive Disorder

Differential diagnosis

It is essential to consider a thorough (neuropsychiatric) differential diagnosis when dealing with cases of depression, given its varied symptomatic presentation. For instance, patients with depression may not predominantly present with low mood and irritability; agitation and somatic symptoms may be more pronounced or be the main presenting complaints.[\[50\]](#) The common differentials to be excluded often include bipolar disorder, which is medically relevant so as not to prescribe antidepressants to someone with a history of manic or hypomanic episodes. Indicators of bipolar depression include the presence of psychotic features, marked psychomotor retardation, irritability or anger in

mood, family history of bipolar depression, early onset, and reverse neurovegetative symptoms.

Depression is the fourth step of the grief cycle and is considered to be a natural response to a sense of loss.[\[51\]](#) It should be enquired whether the person is in the process of moving toward the fifth stage of acceptance or has been stuck in this phase for prolonged periods of time. Since there are no universal ways of coping or dealing with stress, the premorbid history becomes important here to understand whether the current response to loss is normal or out of character for the individual patient. It is also necessary to make an informed call here as a clinician to see whether the person indeed needs pharmacological help in managing grief or alternatives like psychotherapy or support groups will suffice. Somatic symptoms are the most commonly reported symptoms in the clinical setting, and people may often visit general physicians, gastroenterologists, or other specialists for the symptoms that they are experiencing. Identifying the primary or underlying condition might be essential for adequate treatment. Underlying medical issues may also cause secondary depression or may present with symptoms similar to depressive episodes. Identifying depression in individuals with comorbid medical illnesses is also challenging. Additionally, a number of medications may induce depressive symptoms, resulting in medication-induced depression.

MANAGEMENT PRINCIPLES

Goals of treatment

- Achieve symptomatic remission and ensure patient safety and treatment acceptability (acute phase)
- Prevent relapse (continuation phase)
- Prevent recurrent, restore functioning and quality of life (maintenance phase)

Formulation of treatment plan

It involves deciding on the treatment setting, medications, and psychological treatments to be used. Various treatment phases during treatment of depression are explained in [Table](#)

6. Patients and caregivers may be actively consulted while preparing the treatment plan that can be continuously re-evaluated and modified as required. In general, the rule of thumb is that patients may be treated in the setting that is least restrictive yet safest and most effective. Most mild-to-moderate depression cases can be managed in outpatient settings. Inpatient care may be indicated for:

Table 6.

Treatment phases

Phase	Duration	Objectives
Acute	8–16 weeks	Address safety concerns Select evidence-based treatment Achieve symptom remission
Continuation	4-9 months	Consolidate remission; Prevent relapse
Maintenance	6–24 months	Restore functioning and quality of life Prevent recurrence
Recurrent depression	Long term	Prevent recurrence

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- Severe depression with psychosis
- Depression with suicidal or homicidal behavior
- Refusal to eat/severe malnutrition
- Depression with catatonia
- Depression with physical/psychiatric comorbidities make outpatient care unsafe.

All inpatients should preferably have accompanying family caregivers. In case inpatient care facilities are not available, then the patient and/or family need to be informed about

the indication for admission, and the admission in the nearest available inpatient facility can be facilitated. Additionally, close follow-up and strict supervision at home by family may also be a pragmatic approach.

Key Neurobiological Mechanisms of Antidepressant Action

- Antidepressants initially increase synaptic availability of monoamines (e.g., serotonin, noradrenaline), but this alone does not explain clinical response.
- Sustained treatment induces downstream molecular changes, including epigenetic modulation of gene expression.
- These changes enhance neurotrophic signalling (e.g., BDNF), supporting neuronal survival and resilience.
- Antidepressants promote neuroplasticity by facilitating synaptic remodelling and strengthening neural connectivity in mood-regulating circuits.
- Structural and functional recovery occurs in key regions implicated in depression, such as the prefrontal cortex and hippocampus.
- The typical 2–4 week delay in clinical response reflects the time required for gene-level, synaptic, and network-level adaptations.
- Antidepressants act by gradually modulating neural circuits rather than producing immediate symptomatic relief through neurotransmitter changes alone.

It is also worth noting that persistent depressive symptoms are common even in individuals who otherwise achieve symptom remission. These “residual” symptoms should be identified and treated because persisting symptoms represent a major risk factor for recurrence.

INITIAL TREATMENT SELECTION

The choice of initial treatment for MDD should be made collaboratively between the clinician and patient [[Table 7](#)].

Table 7.

Initial treatment selection based on the severity of depression

Severity	Recommendation for initial treatment selection
Mild with low safety risk	• Psychotherapy (if readily accessible or acceptable to the patients) can be preferred because of fewer risks. • Psychotherapy or pharmacotherapy may demonstrate similar benefits. However, in the Indian context, medication can be given if required/preferred by the patient/indicated or due to the lack of readily available psychologists (refer to psychotherapy in depression section). • Furthermore, medication can be preferred as it acts quickly, along with alternative therapies and lifestyle interventions, and supportive interventions like externalization of interests • Exercise, certain complementary and alternative medicine (CAM) treatments, or guided digital health interventions (DHIs) may be considered as monotherapy if preferred by patients.
Moderate with low-moderate safety risk	• The initial choice is between pharmacotherapy and psychotherapy • Pharmacotherapy is slightly more efficacious and gives quick relief in reducing acute symptoms and is cost-effective. • Structured psychotherapy, such as cognitive behavioral therapy (CBT), is efficacious in the medium term (6–12 months), but its availability is the main problem. It is available only in large cities or big treatment centers. The majority of patients

cannot access or even afford the time and money required (refer to psychotherapy in depression section).

- A combination of the above can be used
- Exercise, certain CAM treatments, and/or DHIs may be considered as adjuncts to psychotherapy and/or pharmacotherapy if preferred by patients
- For severe depressive episode without psychotic symptoms, use a combination of pharmacotherapy and psychotherapy (psychotherapy is very difficult or impractical in severe depression in the acute phase. It can only be instituted after some improvement in symptoms) (refer to psychotherapy in depression section)
- For a severe depressive episode with psychotic symptoms, a combination of antidepressant and antipsychotic medication can be used as the first line of treatment.
- For life-threatening situations, consider electroconvulsive therapy and hospitalization

Severe with moderate to high safety risk

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Antidepressant selection

When initiating pharmacotherapy for depression, clinicians can choose from available antidepressants, of which many are considered first-line options based on robust randomized placebo-controlled trials evidence supporting their safety and efficacy [[Table 6](#)].[\[6,52\]](#) A comprehensive network meta-analysis of placebo-controlled and active-comparator trials confirmed that all these agents are more effective than placebo.[\[53\]](#)

The initial selection should be made considering the factors mentioned in [Table 6](#). Before

prescribing, clinicians should provide psychoeducation—discussing expected benefits, possible risks, anticipated time to effect, and addressing myths or misconceptions about antidepressant use. Specific clinical and patient-related factors that influence antidepressant choice are outlined in [Table 8](#). Details of mechanism and dose ranges of commonly used antidepressants are presented in [Table 9](#).

Table 8.

Clinical and patient factors influencing antidepressant choice

Clinical Profile	Comorbidity	Medication-Related Factors	Other Factors
• Clinical features such as loss of appetite/increased appetite, decreased sleep/oversleep (clinical specifiers as per DSM-5) • Age • Pregnancy/postpartum • Previous drug response • Family history of drug response	• Psychiatric comorbidities • Comorbid physical illnesses	• Drug efficacy • Drug tolerability (side-effects) • Potential drug–drug interactions • Ease of availability • Cost-effective	• Patient preference • Doctors’ preference

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(DSM-5: Diagnostic and Statistical Manual of Mental Disorders, 5th edition)

Table 9.

Commonly recommended antidepressants: Mechanism and dose ranges

Antidepressant	Mechanism of Action	Dose Range
First-line antidepressants		
Escitalopram	SSRI	10–20 mg
Sertraline	SSRI	50–200 mg
Fluoxetine	SSRI	20–60 mg
Fluvoxamine	SSRI	100–300 mg
Paroxetine	SSRI	20–50 mg; 25–37.5 mg for CR version
Citalopram	SSRI	20–40 mg
Vortioxetine	Serotonin reuptake inhibitor; 5-HT _{1A} agonist; 5-HT _{1B} partial agonist; 5-HT _{1D} , 5-HT _{3A} , and 5-HT ₇ antagonist	10–20 mg
Duloxetine	SNRI	40–60 mg
Desvenlafaxine	SNRI	50–100 mg
Venlafaxine	SNRI	75–225 mg
Milnacipran*	SNRI	100 mg
Bupropion	NDRI	150–300 mg
Mirtazapine	α 2-Adrenergic antagonist; 5-HT ₂ antagonist, noradrenergic and	15–45 mg

	specific serotonergic antidepressant (NaSSA)	
Mianserin*	α 2-Adrenergic agonist; 5-HT2 antagonist	60–120 mg
Agomelatine	MT1 and MT2 agonist; 5-HT2 antagonist	25–50 mg

Second-line antidepressants (higher side effect burden)

Amitriptyline, clomipramine, and others	TCA	Various dosages
Levomilnacipran *(lack of comparative and relapse-prevention data)	SNRI	-
Moclobemide*	Reversible inhibitor of MAO-A	300–600 mg
Selegiline transdermal*	Irreversible MAO-B inhibitor	6–12 mg daily transdermal
Trazodone	Serotonin receptor antagonist and reuptake inhibitor (SARI): Serotonin reuptake inhibitor; 5-HT2 antagonist	150–300 mg
Vilazodone (lack of comparative and relapse prevention data, need to titrate and taken with food) [DSG16]	Serotonin reuptake inhibitor; 5-HT1A partial agonist	20–40 mg (titrate from 10 mg)

Third-line antidepressants (higher side effect burden, potential serious drug, and dietary interactions)

Phenelzine*	Irreversible MAO inhibitor	45–90 mg
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Tranylcypromine	Irreversible MAO inhibitor	20–60 mg
Reboxetine (lower efficacy) *	Noradrenaline reuptake inhibitor	8–10 mg

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(5-HT: 5-Hydroxy Tryptamine (Serotonin); MAO: monoamine oxidase; MT: melatonin; NDRI: noradrenaline and dopamine reuptake inhibitor; SNRI: serotonin and noradrenaline reuptake inhibitor; SSRI: selective serotonin reuptake inhibitor; TCA: tricyclic antidepressant). *Subject to availability in India

See Box 1 for details of mechanisms underlying the therapeutic effects of antidepressants.

All antidepressants have the potential to cause adverse drug reactions (ADRs) with profiles varying both across and within classes. Selective Serotonin Reuptake Inhibitors (SSRIs) and Selective Norepinephrine Reuptake Inhibitors (SNRIs) are most often associated with gastrointestinal disturbances and sexual dysfunction, whereas tricyclic antidepressants (TCAs) are more likely to produce anticholinergic effects such as dry mouth and constipation. Important side effects such as sedation, postural hypotension, weight gain, and sexual dysfunction are listed in [Table 10](#). These should be discussed with patients before initiation and reassessed within 2 weeks. Baseline investigations' findings can inform clinicians about the suitable psychotropic agents for the patients.

Table 10.

Sedation, weight gain, and insomnia potential of antidepressants[[54](#)]

Antidepressant	Sedation	Weight Gain	Insomnia
SSRIs			
Fluoxetine	0-1	0-1	2-3

Sertraline	0-1	0-1	2
Paroxetine	1-2	2-3	1
Escitalopram	0-1	0-1	1-2
Fluvoxamine	1-2	0-1	1-2
SNRIs			
Venlafaxine	0-1	0-1	2-3
Desvenlafaxine	0-1	0-1	2-3
Duloxetine	0-1	0-1	2
Others			
Mirtazapine	3-4	3-4	0
Bupropion	0	0	3-4
Agomelatine	0-1	0	1
Vortioxetine	0-1	0-1	1
Tricyclics (TCAs)			
Amitriptyline	3-4	3-4	0
Clomipramine	2-3	2-3	0-1
Nortriptyline	2-3	2-3	0-1

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(*0=minimal/absent, 1=mild, 2=moderate, 3=moderately high, 4=high)

For most adults with MDD, the initial choice of antidepressant is guided by the balance between efficacy and tolerability. Evidence-based treatment recommendations across specifiers and clinical dimensions in MDD are summarized in [Table 11](#). A large network meta-analysis found that escitalopram, vortioxetine, and agomelatine ranked higher than others when both efficacy and acceptability were considered. Eight antidepressants bupropion, escitalopram, mirtazapine, paroxetine, sertraline, venlafaxine-XR, vortioxetine,

and agomelatine showed superior response rates compared to others, although the differences were modest (5–10% above comparators), and acceptability, measured by all-cause discontinuation, was also more favorable for several these agents.[\[53\]](#)

Table 11.

Evidence-based treatment recommendations across specifiers and dimensions in MDD[\[6\]](#)

Specifiers/dimensions	Recommendations (level of evidence)	Comments
With anxious distress	• Use an antidepressant with efficacy in generalized anxiety disorder (Level 4)	• No differences in efficacy between SSRIs, SNRIs, and bupropion (Level 2)
With catatonic features	• Benzodiazepines (Level 3)	• No antidepressants have been studied
With melancholic features	• No specific antidepressants have demonstrated superiority (Level 2)	• TCAs and SNRIs have been studied
With atypical features	• No specific antidepressants have demonstrated superiority (Level 2)	• Older studies found MAO inhibitors superior to TCAs
With psychotic features	• Use antipsychotic and antidepressant cotreatment (Level 1)	• Few studies involved atypical antipsychotics
With mixed features	• Lurasidone (Level 2)	• No comparative studies

	• Ziprasidone (Level 3)	
With seasonal patterns	• No specific antidepressants have demonstrated superiority (Level 2 and 3)	• SSRIs, agomelatine, bupropion, and moclobemide have been studied
With cognitive dysfunction	• Vortioxetine (Level 1) • Bupropion (Level 2) • Duloxetine (Level 2) • SSRIs (Level 2) • Moclobemide (Level 3)	• Limited data available on the cognitive effects of other antidepressants and on comparative differences in efficacy
With sleep disturbances	• Mirtazapine (Level 2) • Quetiapine (Level 2) • Trazodone (Level 2) • Agomelatine (Level 1)	• Beneficial effects on sleep must be balanced against the potential for side effects (e.g., daytime sedation)
With somatic symptoms	• Duloxetine (pain) (Level 1) • Other SNRIs (pain) (Level 2) • Bupropion (fatigue) (Level 1) • SSRIs (fatigue) (Level 2) • Duloxetine (energy) (Level 2)	• Few antidepressants have been studied for somatic symptoms other than pain • Few comparative antidepressant studies for pain and other somatic symptoms

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(MAO: monoamine oxidase; SNRI: serotonin and noradrenaline reuptake inhibitor; SSRI: selective serotonin reuptake inhibitor; TCA: tricyclic antidepressant). Notes: aDSM-5 specifiers. Comparisons only with placebo. Level of evidence: (1) Meta-analysis with narrow confidence intervals and/or 2 or more randomized controlled trials (RCTs) with adequate sample size, preferably placebo controlled; (2) meta-analyses with wide confidence intervals and/or 1 or more RCTs with adequate sample size; (3) small-sample RCTs or nonrandomized, controlled prospective studies or case series or high-quality retrospective studies; and (4) expert opinion/consensus

Other considerations include drug–drug interactions as fluoxetine, fluvoxamine, and paroxetine have clinically relevant potential interactions. Citalopram can prolong the QTc interval (it is not available in India); escitalopram does not pose a clinically significant risk of QTc prolongation. Most antidepressants can elevate liver enzymes, but agomelatine specifically requires regular liver function monitoring.

Sexual dysfunction, which may also be a symptom of depression, should be assessed at baseline to distinguish pre-existing from treatment-emergent effects. Desvenlafaxine, bupropion, mirtazapine, vilazodone, vortioxetine, and agomelatine are associated with lower rates of sexual side effects. Age may also influence choice: In older adults, SSRIs may be less effective, and SNRIs such as duloxetine may be more suitable, while in patients aged 25 or younger, fluoxetine or agomelatine may offer a better balance of efficacy, tolerability, and risk of discontinuation symptoms.

DURATION OF ANTIDEPRESSANT TREATMENT

After starting an antidepressant, patients should be monitored closely for both treatment response and side effects. The frequency of monitoring depends on the severity of illness, level of suicide risk, comorbid medical conditions, medication tolerability, social support, and treatment compliance. Timely follow-up visits help ensure safety, detect emerging side effects or suicidality, and promote adherence. Antidepressants should be started at a lower dose and titrated gradually based on response and tolerability. Clinical improvement typically appears after 4–6 weeks of adequate treatment.

- If $\geq 25\%$ improvement is observed after 4 weeks, continue and optimize to the maximum tolerated dose.
- If $< 25\%$ improvement is observed after 4 weeks, reassess adherence, interactions, and diagnosis and consider changing the antidepressant.
- If $< 50\%$ improvement is observed even after 6–8 weeks of treatment with optimal dose with good adherence, modify treatment by either switching or augmenting.

If no significant improvement is observed after 4–8 weeks, a comprehensive review of the diagnosis, medication compliance, comorbidities (both medical and psychiatric), and psychosocial stressors should be conducted. The treatment plan can be revised considering one of the following strategies:[\[55\]](#)

- Maximize the current antidepressant dose.
- Switch to another antidepressant (within or across pharmacologic classes).
- Augment with another medication (e.g., lithium, thyroid hormone, second-generation antipsychotics, combination of antidepressant medications, dopaminergic agonists, etc.) or combine with psychotherapy or electroconvulsive therapy (ECT).

Continuation phase

Patients who respond to acute treatment should continue the same dose for 16–24 weeks (total duration ~ 6 –9 months from initiation) to prevent relapse. Evidence supports combining specific psychotherapies (e.g., Cognitive Behavior Therapy, Interpersonal Therapy during the continuation phase to enhance relapse prevention (more details provided in the psychotherapy for depression section)). Somatic treatments (e.g., ECT) may be used when medications or psychotherapy fails to maintain stability. Follow-up visit frequency should be individualized according to clinical condition and treatment modality. Patients with recurrent and severe depressive episodes should use sequential treatment (adding psychotherapy after stabilizing patients on medications) to prevent recurrence.

Maintenance phase

It is recommended to continue antidepressant treatment for 6–12 months after full remission. Patients with risk factors for recurrence (e.g., recurrent or severe or chronic depressive episodes, history of childhood maltreatment, comorbid anxiety or physical comorbidity, poor social support, persistent stressful life events or residual symptoms) should remain on maintenance therapy for ≥ 2 years. Sequential treatment, that is, adding psychotherapy after stabilization on medication, further reduces relapse risk.

DISCONTINUATION OF TREATMENT

The decision to discontinue maintenance treatment may be based on the same factors considered when initiating maintenance treatment, including the probability of recurrence, the frequency and severity of past episodes, the persistence of depressive symptoms after recovery, the presence of comorbid disorders, and patient preferences. However, there is no consensus on the best strategy for stopping antidepressants. When a decision is made to stop the antidepressant, it should be tapered gradually, whenever possible, for several weeks or months, with more time between dose reductions near the end of the taper. Psychological treatments can be added before or during antidepressant discontinuation to help patients stop the antidepressant and consolidate on the gains brought by the medications. In addition, such tapering can help minimize the risks of antidepressant medication discontinuation syndromes. Discontinuation syndromes have been found to be more frequent after discontinuation of medications with shorter half-lives, and patients maintained on short-acting agents may be given even longer, more gradual tapering. Paroxetine, fluvoxamine, and venlafaxine tend to have higher rates of discontinuation symptoms, while bupropion-SR, fluoxetine, mirtazapine, vortioxetine, and agomelatine have lower rates. The symptoms of antidepressant discontinuation include flu-like symptoms, insomnia, nausea, imbalance, sensory disturbances (e.g., electrical sensations), and hyperarousal (agitation).[\[56,57\]](#) If the discontinuation syndrome is mild, reassurance may be sufficient. If mild to moderate, short-term symptomatic treatment (analgesics, antiemetics, or anxiolytics) may be beneficial. If it is severe, the antidepressant must be reinstated and tapered off more slowly.

After the discontinuation of active treatment, patients should be reminded of the potential for a depressive relapse. The patient should be psychoeducated about the early warning signs of relapse. If a patient suffers a relapse upon discontinuation of medication,

treatments need to be promptly reinitiated. In general, the previous treatment regimen to which the patient responded in the acute and continuation phase is to be considered.

Management of comorbidity

Comorbidity with psychiatric and medical conditions is common in MDD and significantly affects management. The presence of comorbid conditions often makes MDD more challenging to treat, and differentiating the side effects of depression treatment from those of other medications can be difficult. Minimizing polypharmacy, screening for drug–drug interactions, and selecting antidepressants with favorable side effect profiles are essential considerations.

Patients with psychiatric and medical comorbidities tend to have a poorer response to antidepressants than those without, yet pharmacotherapy remains indicated. An umbrella systematic review and meta-analysis of RCTs in MDD with 27 comorbid medical conditions found antidepressants to be more effective than placebo across a range of illnesses, including myocardial infarction, coronary artery disease, stroke, inflammatory bowel disease, diabetes mellitus, and cancer. Psychological treatments are also effective in MDD with comorbidities. Despite study heterogeneity, meta-analyses support the use of CBT and behavioral activation (BA) for patients with specific medical comorbidities such as coronary heart disease and neurological disorders.

SPECIAL POPULATIONS

- **Perinatal depression**—Avoid teratogenic medications; consider psychotherapy as first-line in mild cases; SSRIs with established safety profiles (sertraline) if pharmacotherapy is needed.
- **Elderly**—Start low, go slow with medications; monitor for side effects, especially hyponatremia and falls. Usually requires half of the starting and targeting dose than adults.

TREATMENT-RESISTANT DEPRESSION AND DIFFICULT-TO-TREAT DEPRESSION

Treatment-resistant depression (TRD) commonly refers to failure to respond to at least two antidepressants at an adequate dose and duration, though definitions vary and often overlook psychological or neurostimulation treatments, partial responses, and add-on strategies.^[58] The term “resistant” may carry negative connotations that discourage ongoing treatment. Owing to these conceptual issues with TRD, an alternative term was proposed: “Difficult-to-treat depression (DTD).” The consensus definition of DTD, or suspected DTD, is depression that continues to cause significant burden despite usual treatment efforts.^[59] In DTD, the focus shifts from full remission to optimizing functioning, quality of life, and patient-prioritized outcomes. In the current guidelines, “DTD” is used as the holistic framework, while “TRD” is retained when referring to medication options or studies that use that definition.^[59]

Assessment before confirming TRD/DTD

- Reassess diagnosis (rule out bipolar disorder, comorbidities)
- Check treatment adherence
- Review adequacy of past trials (dose, duration, switching strategy)
- Identify medical causes (e.g., hypothyroidism, vitamin deficiencies)
- Evaluate psychosocial stressors

Recommended sequential interventions for difficult-to-treat depression are presented in [Table 12](#).

Table 12.

Recommended sequential interventions for difficult-to-treat depression^[59,60,61,62,63]

Step	Intervention	Notes
Optimize current treatment	• Increase to the maximum tolerated dose	• Monitor ADRs

Switch antidepressant	• Use a different mechanism of action	• E.g., SSRI → SNRI; SNRI → bupropion
Augmentation	• Add lithium, thyroid hormone, or an atypical antipsychotic	• Greatest evidence available for lithium and aripiprazole
Combination therapy	• Combine two antidepressants from different classes	• Use with caution
Neuromodulation	• Electroconvulsive therapy (ECT) (noninvasive), repetitive transcranial magnetic stimulation (rTMS) (noninvasive)	• ECT for severe/suicidal cases
Novel agents	• Ketamine/esketamine, dextromethorphan-bupropion (wherever available)	• Limited long-term data

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ADRs: adverse drug reactions; ECT: electroconvulsive therapy; rTMS: Repetitive Transcranial Magnetic Stimulation; SSRI: selective serotonin reuptake inhibitor; SNRI: serotonin-norepinephrine reuptake inhibitor

ADRs: adverse drug reactions; ECT: electroconvulsive therapy; rTMS: Repetitive Transcranial Magnetic Stimulation; SSRI: selective serotonin reuptake inhibitor; SNRI: serotonin-norepinephrine reuptake inhibitor

NOVEL PHARMACOLOGICAL AGENTS IN MDD

Ketamine

Intranasal esketamine administered alongside an antidepressant was approved by the U.S. FDA in March 2019 for adults with TRD, and the European Medicines Agency approved it in December 2019. In August 2020, it also received FDA approval for patients with MDD experiencing suicidal ideation or behavior.[\[64\]](#) Routes and dosage used are presented in [Table 13](#).

Table 13.

Route and dosage of Ketamine[[65](#)]

Route	Dose range	Method of administration
Intravenous ketamine	0.5–1.0 mg/kg	• Infused over 40–60 minutes, twice weekly for 2 weeks
Oral ketamine	2–3 mg/kg	• Required quantity mixed with around 100 mL of purified water (flavors may be added); sipped over 10–15 minutes for the first session and a little more rapidly in subsequent sessions (<i>The available evidence for oral ketamine is preliminary and currently insufficient to support routine clinical use.</i>)
Sublingual	Not established, however, to start with 10 mg	• To place under the tongue without swallowing for 2 minutes
Intranasal	• Esketamine: 56–84 mg intranasally twice weekly for 4 weeks • Ketamine: 50–150 mg intranasally twice weekly	• Nasal spray (comes with a proprietary dispenser that administers fixed doses)

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Indications:[[65](#),[66](#),[67](#)]

- TRD
- MDD with suicidal thoughts/behavior
- Treatment-resistant bipolar depression
- Augmentation with ECT (small studies)

Adverse effects with ketamine

- **Cardiovascular:** Increased heart rate and blood pressure, arrhythmias (monitor vitals)
- **Psychiatric:** Dissociation, perceptual changes, anxiety, rare psychosis or delirium
- **Neurological:** Headache, dizziness, confusion
- **Others:** Nausea, dry mouth, blurred vision, increased intraocular pressure, dysphoria, anxiety, dissociative experiences (unusual thoughts, dream-like feeling of unreality, unawareness of self and environment, illusions, and hallucinations), treatment-induced affective switch, precipitation of psychosis, and rarely emergence of delirium.

Monitoring:[68]

Continuous monitoring of consciousness, heart rate, blood pressure, respiratory rate, and oxygen saturation during baseline and periodically (e.g., every 15 minutes) throughout the session and continuing for up to 2–3 hours after termination of the session.

Instruments such as the Clinician Administered Dissociative States Scale (CADSS) can be used for a structured assessment of dissociative/psychotomimetic experiences.

Treatment is typically administered in the form of six to eight infusions, given twice or three times weekly (acute phase), to prolong the gains. In some cases where this is a nonresponse to other lines of treatment, ketamine therapy may also be given as a continuation and maintenance phase with individualized frequency, although most of the evidence on these lines is anecdotal.

Dextromethorphan–bupropion combination – Approved in USA (2022) for MDD and now available in India also. Dextromethorphan is a glutamate NMDA receptor antagonist,

sigma-1 receptor agonist, and SNRI, also used as a cough suppressant. Bupropion inhibits CYP2D6, increasing dextromethorphan bioavailability, and acts as a dopamine/norepinephrine reuptake inhibitor/releaser. Recommended as **second-line** due to limited evidence, potential misuse risk, and lack of long-term safety data.

Brexanolone – An allopregnanolone agonist, approved in USA for postpartum depression.

Zuranolone – An oral allopregnanolone agonist, also approved in USA for postpartum depression.

Psychotherapy for depression

Psychological intervention or psychotherapy (henceforth, psychotherapy) is one of the essential components of depression treatment. In fact, they are one of the first-line treatments for mild to moderate depression as per many guidelines.[\[69,70,71\]](#) Like other mental health conditions, depression and its treatment involve a bio-psycho-social approach to management. Psychosocial factors can serve as the precipitating factor for depression (e.g., recent loss of loved ones, financial loss, loss of a job), predisposing factor (e.g., personality factors, unhealthy coping techniques, poor social support, adverse childhood experiences, history of trauma), and perpetuating factors (e.g., interpersonal issues, financial or health issues). On the other hand, there could be various protective factors against depression that can be utilized during psychotherapy for depression (e.g., family or social support, religious and cultural beliefs)[\[69,70,71\]](#) [\[Table 14\]](#).

Table 14.

General considerations for psychotherapy in depression

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- Psycho-educating the patient/family members about depression, involving bio-psycho-social model, is an essential aspect of psychotherapy.
 - Psychotherapy should focus on the personalized care for individuals with depression, considering their unique needs and preferences.

- Alliance in individual psychotherapy (with a collaborative stance on goals, therapeutic tasks), empathy, and monitoring treatment response through standardized scales have proven effectiveness.
- Emphasis should be placed on treatment adherence, including addressing factors linked to it.
- Psychotherapy is suitable for depressed patients across all ages, genders, educational levels, and cultural and ethnic backgrounds.
- Time course of improvement is faster and more sustained when pharmacotherapy is used along with psychotherapy. The same should be conveyed to the patients.
- The frequency of sessions in the acute or maintenance phase of the treatment can vary depending upon the psychotherapy or patient's condition. However, more frequent sessions in a week during the early phase of the treatment have greater effectiveness.
- Brief therapies (≤ 8 sessions of CBT, IPT, Problem-solving therapy) can be effective; however, concerning the number of sessions, specific therapy modules should be followed to ensure effectiveness.
- Psychological therapies for depression should be delivered by trained and proficient therapists. Although less trained therapists are also effective and at times the only available options.
- Collecting patients' self-reported outcomes and objective measures for treatment progress, and obtaining feedback from colleagues/supervisors during psychotherapy, improves clinical outcome.
- Patients' characteristics (psychosocial characteristics, patients' preference for type, format and therapists' characteristics, e.g., gender) need to be considered while selecting or delivering psychotherapy
- Availability, cost, and accessibility are critical considerations while planning and delivering psychotherapy
- Severity of depression, prominent symptoms and the associated comorbidities (physical or psychiatric illness) must be accounted for while delivering psychotherapy

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Like pharmacotherapy, administration of psychotherapy should be based on patients’ characteristics, preferences, availability, accessibility, and affordability. There should be shared decision-making between the patient and the therapist for planning psychotherapy; moreover, the role of family factors, including support for regular therapy sessions and adherence to treatment, needs to be carefully considered. Literature shows personalized psychotherapy for depression involving prioritizing patients’ experiences and preferences, a facilitatory approach, focus on psychoeducation, reinforcing therapeutic alliance, and promoting patients’ agency, has better outcomes.[71,72]

THE CURRENT EVIDENCE FOR PSYCHOTHERAPY IN DEPRESSION

Psychotherapy is a vital treatment modality both in the acute phase of depression (remission of symptoms) and in the continuation/maintenance phase (preventing relapse/recurrence). In general, a combined treatment (pharmacotherapy with psychotherapy) should be offered to patients with moderate to severe depression. However, this should be based on weighing the benefits and burdens, as well as patients’ preferences.[69,70,71] Research shows that the combined treatment is more effective than pharmacotherapy or psychotherapy alone for depression, in both acute and maintenance phases.[73]

American Psychological Association, 2019 (APA, 2019) guidelines states that behavioral therapy, cognitive, cognitive-behavioral, and mindfulness-based cognitive-therapy, interpersonal psychotherapy, psychodynamic therapies, and supportive therapy have comparable efficacies in improving depressive symptoms compared to the active control. Therefore, it recommends use of above psychotherapies in depression. However, slightly older national guidelines (Canadian Network for Mood and Anxiety Treatments [CANMAT], 2016) recommend cognitive behavioral therapy (CBT), interpersonal therapy (IPT), and behavior al activation as a first-line therapy for the *acute phase* of depression, either standalone (for mild to moderate depression) or in combination with antidepressants (severe or moderate depression)[69,71,74] [Tables 15 and 16].

Table 15.

Psychotherapies and their evidence for different phases of depression treatment

Recommendation	Strength of Recommendation	Justification
Acute phase treatment		
For initial treatment, following the context of shared-decision process, following options can be considered: Clinicians can offer either psychotherapy or second-generation antidepressants: • Second-generation antidepressants • Psychotherapy with comparable efficacies - Behavior al therapy - Cognitive, cognitive-behavioral, and mindfulness-based cognitive-therapy - Interpersonal psychotherapy - Psychodynamic therapies - Supportive therapy	Recommended treatment	• In case psychotherapy is not available – due to lack of resources or unavailability of trained psychotherapist, or individuals have preference for pharmacotherapy, second-generation (SSRIs or SNRIs) antidepressants (ADs) should be considered. • MB-CT is particularly useful for residual depressive symptoms. Also, specifically helpful for individuals with greater vulnerability (e.g., recurrent depression, unstable remission, or a history of childhood trauma).
If considering combined therapy- a clinician can offer CBT or IPT and second-generation antidepressants	Recommended treatment	• Combined treatment has shown better efficacy, greater treatment adherence,

depression. However, clinicians should weigh benefit–burden ratio of combined treatment

Cognitive behavioral analysis system of psychotherapy (CBASP), Brief problem-solving therapy (brief-PST) (10 or fewer sessions) vs treatment as usual.

Insufficient evidence for a recommendation

Depression with relational distress:
If recommended treatment (antidepressants and other recommended therapies mentioned above) is not acceptable or available, consider problem focused couple's therapy, treatment

Conditional recommendation for use

and early remission.

- However, patients' preference, and cost of the care and logistics should be taken into consideration while offering such intervention
- Research consistently shows that short-term psychodynamic (STPP) (not long-term psychodynamic psychotherapy [PDP]), makes outcome worse in some domains compared to standard psychotherapies.
- CBASP, brief-PST in depression has insufficient evidence on effectiveness in depression over treatment as usual (TAU).
- Treatment needs to be tailored based on individuals' characteristics.
- Marital or relational distress do not simply respond to standard pharmacotherapies.
- Improvement in

options include:
Behavioral therapy over AD
alone
Combining CBT with AD
improves likelihood of complete
remission

partner relationships
may favor couples
therapy (or
incorporating it as a
component in standard
therapy) when
relational distress is a
major problem

Partial responders or non-responders to initial antidepressants treatment

Switch from antidepressants to above recommended psychotherapy alone OR, Switch to another AD	Recommended for use	Evidence suggests there is no difference in the efficacy of the second- generation antidepressants and standard psychotherapy (CBT, IPT, BT, PDP)
Add psychotherapy (IPT, CBT, or PDP) to the existing AD treatment Augment with another AD	Conditional Recommendation for use	Addition of CBASP, brief-supported psychotherapy to AD do not yield additional benefit.

Maintenance treatment/relapse prevention

Individuals who have achieved remission with the initial treatment, to prevent relapse, psychotherapy (CBT, MB-CT, IPT) to be offered over AD alone or continuing with treatment as usual.	Conditional recommendation for use	Psychotherapy in general (CBT, MB-CT, IPT) demonstrated comparable efficacy to AD or TAU to prevent relapse. There is insufficient evidence to recommend one psychotherapy over others.
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*Treatment should always involve shared decision-making. ADRs: adverse drug reactions; CBASP - cognitive behavioral analysis system of psychotherapy, CBT – cognitive behavioral therapy, IPT – interpersonal therapy, MB-CT – mindfulness-based cognitive therapy, MHPs – mental health professionals, PDP – psychodynamic psychotherapy, PDT – Long-term psychodynamic psychotherapy, psychotherapy, PST – problem solving therapy, SNRIs – serotonin, norepinephrine reuptake inhibitor, SSRIs - selective serotonin reuptake inhibitor, STPP – short-term psychodynamic psychotherapy

Table 16.

Psychotherapy for suicidality in depression level of evidence and critical aspects

Psychotherapy	Level of evidence/level of recommendation for prevention of SI	Level of evidence/level of recommendation for prevention of SAs	Critical aspect
CBT with components tailored to suicide prevention and problem-solving approach	Level 1 evidence; therefore, recommended for use along with antidepressants for moderate to severe depression without psychotic symptoms and suicidal ideations	Level 1 evidence; therefore, recommended for use along with antidepressants for moderate to severe depression without psychotic symptoms and suicidal ideations	• The core CBT components (targeting the maladaptive cognitive processes) are often supplemented by problem-solving skills, interpersonal

skills, emotional regulation skills, and safety planning to prevention suicidal behavior in depressed individuals.

DBT with problem-solving approach	Level 1 evidence; therefore, recommended for use along with antidepressants for moderate to severe depression without psychotic symptoms and suicidal behavior.	Level 1 evidence; therefore, recommended for use along with antidepressants for moderate to severe depression without psychotic symptoms and suicidal behavior.	• DBT adapted for the depression with components of the CBT, problem-solving, interpersonal skills, and mindfulness-practices provides a comprehensive package to manage suicidal behavior in depressed individuals
IPT	Level 1 evidence; conditional recommendation for use with antidepressants	Inadequate evidence to make any recommendations	• IPT, particularly by addressing IP friction, building assertive skills, effective IP communication

skills, bring about improvement in suicide in depressed individuals working through the framework of ideation-to-action model of suicidal behavior.

MB-CT

Level 2 evidence; conditional recommendation for use

Inadequate evidence to make any recommendations

- Particularly, helpful in managing residual depression that can be associated with suicidal behavior.
- The effectiveness is limited to wait-list control or equivalent to TAU.
- Greater evidence is for reducing hopelessness in depression as

Problem Solving Therapy	Level 2 evidence, conditional recommendation for use	Level 1 evidence; therefore recommended for us in depressions with history of suicidal behavior to prevent suicide reattempt (s).	compared to improving suicidality in depression. • The evidence is limited to uncontrolled trials or controlled trials with wait-list control. • However, standalone PST or PST with antidepressants have not been found superior to other standard psychotherapies. • Components of the PST are key elements of other standard psychotherapies in depression with suicidal behavior.
Psychoanalytic or psychodynamic psychotherapy	Level 2 evidence, conditionally recommended treatments	Level 1 evidence, however, due to limited data, it is conditionally recommended treatments for	• Impact lasts up to 6 months. Moreover, lack of well-controlled trials to consider it for

		suicide reattempt prevention in depressed individuals.	the long-term treatment option for suicidal prevention in depression.
Family therapy	Level 3 evidence, Insufficient evidence for a recommendation	Insufficient evidence for a recommendation	• Lack of well-controlled trial limits it's wider usability. Study is limited to the vulnerable community. Therefore, generalizability of the findings is doubtful. • Greater research particularly in depressed individuals with suicidality is required.

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Level of evidence 1 Meta-analysis with narrow confidence intervals and/or 2 or more RCTs with adequate sample size, preferably placebo controlled. Level 2 - Meta-analysis with wide confidence intervals and/or 1 or more RCTs with adequate sample size. Level 3 - Small-sample RCTs or nonrandomized, controlled prospective studies or case series or high-quality retrospective studies. Level 4 - Expert opinion/consensus

The APA Guideline (2019) for psychotherapy in depression recommends that depression with concomitant psychosocial stressors, such as relational distress or marital discord, if the above-mentioned recommended treatments are not available or not acceptable, problem-focused couple therapy should be considered. Moreover, in such conditions, antidepressant monotherapy is not sufficient, and behavioral therapy over antidepressant alone is recommended; combining CBT and AD improves the likelihood of full remission[[69](#)] [[Table 13](#)].

The APA Guideline (2019) recommends that clinicians should offer psychotherapy (CBT, IPT, and mindfulness-based cognitive therapy (MB-CT)) to the individuals who have achieved remission (maintenance treatment/relapse prevention) of depression rather than antidepressants alone or treatment as usual[[69,71,74](#)] (refer Table _ for level of evidence for various psychotherapies and recommendations for use in depression). General considerations while undertaking psychotherapy for depression are provided in [Table 14](#).

With the advent of digital technology and the Internet, computer-based or Internet-based (CB/IB) therapy has become increasingly popular. The CANMAT guideline (2016) reports that the modes of therapy found to be effective include in-person, computer-based, or tele-based, as well as hybrid methods. The therapy format with available evidence supports both individual and group therapy.[[71](#)] The CANMAT (2016) guidelines suggest that CBT or IPT adapted to these modalities of intervention delivery (may be self-guided, guided by a therapist, or asynchronous) is as effective as in-person therapy in the acute phase, as well as for relapse prevention.[[75](#)] Moreover, when CB/IB-therapies are guided by therapists, the adherence to treatment efficacy is substantial.[[76,77,78](#)] However, the data are limited in this regard.[[69](#)] Hence, it is often recommended as a second-line treatment for depression, with support from noninferiority trials.[[71](#)] Likewise, remote live interactive psychotherapy (via phone, video-conferencing) for depression compared to face-to-face therapy has comparable efficacy in depression.[[71,79](#)] Telephone-delivered CBT has Level 1 Evidence (meta-analytical with low confidence interval support), while other types of psychological interventions have Level 2 evidence (supported by ≥ 1 RCTs, with adequate sample size), thus positioning telephone-delivered therapy as a second-line treatment.[[80](#)] Similarly, video-conferencing-based psychological treatments have been found to be acceptable and generally equivalent to face-to-face care for many psychiatric conditions.[[71,81](#)]

- *Recommended* treatments are those consistently shown to be superior to placebo control conditions for which there is evidence of equi-efficacy or superiority over other treatments.
- *Conditionally recommended* treatments that were consistently superior to placebo control conditions, but there were either:
 - Other active treatment(s) are superior to the treatment being conditionally recommended.
 - Insufficient evidence that the treatment was equi-efficacious to other effective treatments.
 - Greater harms/burdens than with other treatments.
- Treatments were not recommended or recommend against if there was insufficient evidence of efficacy or if the harms were considered to outweigh any benefits.
- Insufficient evidence for a recommendation is mentioned
 - If literature is inadequate to make any comments

MANAGEMENT OF SUICIDE IN DEPRESSION

Managing suicidality in depression, like suicide prevention and intervention in general, requires establishing and maintaining a therapeutic alliance with the patients and prioritizing patients' safety. Determining the most suitable treatment setting is paramount. Development of a comprehensive treatment plan, psychoeducation, promotion of treatment adherence, monitoring treatment progress, and undertaking dynamic suicide-risk assessment are key strategies. Choosing evidence-based pharmacological, psychological, and somatic treatment and having a multidisciplinary team on board are the cornerstones of treatment and long-term outcome. Last, but equally importantly, therapists/psychiatrists' own vulnerabilities and their addressal are vital.[\[25,82\]](#) These aspects are discussed in detail below:

Establish and maintain a therapeutic relationship and prioritize patients' safety

A strong therapeutic alliance is critical to suicidal patients' care. Expressing genuine care, empathy, and balancing patients' autonomy with taking appropriate safety measures is

vital. A good therapeutic alliance ensures gathering proper history and gauging future suicidality.[[25](#),[82](#)]

Simultaneously, a doctor/therapist needs to ensure the patient's safety. Whenever required, inpatient care or prolonged observation in the emergency department may be required. It cannot be overemphasized that the patient needs to be continuously monitored, irrespective of the treatment setting, and potential self-inflicting materials (e.g., sharps, cloths, ropes, medications) need to be kept away from the patient. In case of agitated or psychotic patients with self-harm behavior, a physical or chemical restraint can be employed.

Determine a treatment setting

Ideally, the setting of the treatment should be the least restrictive for the patient. However, in case of doubt in the diagnosis or potential suicidal risk, one should not hesitate to admit the patient in the psychiatry facility. The setting of the treatment should also balance patients' safety against the stigma of getting hospitalized or being away from loved ones.

Inpatient setting is useful when there is imminent risk to harm oneself or others; in establishing the diagnoses, including performing psychological assessments; safeguarding a patient against the ongoing or immediate psychosocial stressors; and administering injectables, electroconvulsive therapy (ECT), or ketamine therapy. Also, when outpatient-based treatment fails, an MHP must consider inpatient care for a more elaborate assessment and intervention. It is vital that a psychiatrist assesses the capacity of a patient while considering inpatient care and follows the prevailing mental health law in case of involuntary treatment, including taking consent from the nominated representative or family members.[[83](#)]

OPD-based treatment, however, seems to be a more suitable setting when there is chronic suicidal ideation to which a patient has insight, and they are not intending to end their life or have any plan. A trusted patient–doctor relationship and good social or safety network also indicate an outpatient-based treatment.

Patients and their family members need to be sensitized about the availability of emergency care in case of active suicidality and unmanageability. An MHP must act as a liaison officer or inform the ER care and treatment setting follows a dynamic approach with a stepped care model as per the ongoing condition.

Develop a plan of treatment

A psychiatrist needs to prepare a comprehensive treatment plan after performing an adequate suicide-risk assessment. The treatment may involve biological and psychosocial interventions.

Multidisciplinary care and collaboration with other clinicians

It cannot be overemphasized that care for a person with suicide risk requires a multidisciplinary team approach involving a psychiatrist as the overall in-charge of the case, clinical psychologists, a psychiatry social worker, or a nursing professional as per the availability and need. In such a team-based approach, clear definitions of roles, regular interactions among members, and advanced planning for crisis intervention are essential. During outpatient-based care, an open and regular communication among team members is valuable.

Ensuring and facilitating treatment adherence

Ensuring treatment adherence is paramount for depressed individuals with suicidal behavior, for both the acute and maintenance management. The therapeutic alliance between the clinician and the patients hugely influences treatment adherence. Collaborative decision-making in treatment planning and implementation facilitates adherence to treatment. Moreover, simplifying the treatment regimen and facilitating follow-up or appointments improves adherence, particularly in the elderly. The profound impact of educating on acute phase and maintenance treatment to both patients and caregivers cannot be overemphasized. Importantly, regularly assessing the treatment plan and making course corrections is vital in ensuring treatment adherence.

Psychoeducation

Psychoeducation to patients and family members is a crucial aspect in the management of suicidal behavior in depression. The content of psychoeducation should be informed by clients' unique suicidal phenomena and their psychiatric and psychosocial history. The patient and, with their consent, family members should be educated about the bio-psychosocial model of suicidal behavior. The potential etiology of the suicide behavior, the impact of the psychosocial triggers, treatment options, including expected benefits and potential side-effects of each, past life history, the danger associated with the current suicidal behavior, and the need for long-term support and care form important components of the psychoeducation. The warning signs and symptoms of suicidality (insomnia, agitation, hopelessness, hallucinations, psychosocial stressors, etc.) must be clearly described to patients and family members.

Reassess safety and suicide risk

A regular suicide risk assessment and safety checks should be undertaken by the psychiatrist, particularly when there is a change in the treatment intensity (discharge from the ward, decreasing the frequency of visits, reducing the dose of medications or frequency of therapies), recent change in the symptomatology (worsening in mood symptoms or anxiety, restatement of substance use), or major life changes. These assessments may not be as elaborate as those performed during the initial assessment. Importantly, they should be properly documented, including changes in the treatment plan and the advice given to patients and family members.

Monitor psychiatric status, including depression, and treatment response

Monitoring of current suicide-risk as well as depressive symptomatology is crucial. It is expected that with the improvement in depression or other comorbid mental disorders, suicidality also improves and vice versa, but the relationship is not linear.^[84,85] For instance, the risk of suicidality is higher when the symptoms of depression have started improving, but the depressive cognitions are persisting. Moreover, improvement in

depression does not necessarily mean suicidal behavior will improve by itself; instead, the recovery from suicidal behavior in a depressed individual has its own course and warrants specific interventions.

PHARMACOTHERAPY FOR SUICIDE MANAGEMENT IN DEPRESSION

As described above, the commonly used agents to treat suicidality in depression are antidepressants, mood stabilizers (lithium and anti-convulsant), ketamine, anxiolytics, antipsychotics, and somatic treatment like ECT. Among them, lithium, ketamine, and ECT have the greatest evidence in acutely reducing suicidality.[\[86\]](#) Others may benefit by improving depression *per se* or ameliorating anxiety, agitation, and impulsive urges.

Antidepressants

The primary treatment of suicidal patients with depression is antidepressants. Typically, SSRIs or SNRIs are the first-line treatment. However, they may not independently address suicidality. Apart from improving depression, they also alleviate anxiety, though they can initially worsen arousal symptoms and aggravate suicide risk. Moreover, their effect can only appear after 2–3 weeks. During this period, a psychiatrist can take the help of anxiolytic agents or hypnotics.[\[37\]](#)

Lithium

There is level 1 evidence for the effectiveness of ketamine and lithium in the management of suicidal behavior in depression.[\[38\]](#) Lithium has been shown to improve suicidal ideation in both unipolar and bipolar depression[\[50\]](#) with good tolerability. The improvement is greater than placebo as per the latest Systematic Review and Meta-analysis (SR-MA); however, contrary to previous beliefs, the improvement with other commonly used antidepressants and mood stabilizers, such as amitriptyline, carbamazepine, lamotrigine, and olanzapine, respectively, is not statistically different from placebo.[\[38\]](#)

Ketamine

Ketamine infusion in sub-anesthetic doses (0.5 mg/kg body weight) has been shown to improve suicidal ideations in depressed individuals with minimal adverse drug effects. The reported ADR is hypertension during the time of induction; however, it is self-remitting. [\[51\]](#)

Anxiolytic drugs

There is no standard recommendation to co-prescribe anxiolytics; however, they help address the agitation, panic, insomnia, and anxiety, especially during the initial phase of treating depression with suicidal behavior. Benzodiazepines or other anxiolytics can be used for a short term (1–4 weeks) only and should be gradually tapered off. Usually, a longer active agent is preferred (e.g., clonazepam) as shorter-acting agents can have rebound anxiety in between, which then can worsen agitation. Long-term benzodiazepine use also carries a risk of dependence and subsequent withdrawal symptoms, which can, in fact, lead to suicidal risk. This must, however, be borne in mind that benzodiazepines can precipitate disinhibition or aggressiveness, especially in persons with borderline disorders; hence, they should be used with caution.

Newer agents such as gabapentin or pregabalin or anticonvulsants (e.g., divalproex) seem to be better options to manage psychic anxiety or agitation in persons with suicidal behavior. Moreover, to manage insomnia in depressed individuals with suicidal risk, sedative antidepressants, such as mirtazapine, trazodone, or TCAs, can be used. Other potential options are second-generation antipsychotics (e.g., quetiapine or olanzapine in low dose). [\[37\]](#)

Electroconvulsive therapy

ECT has a strong empirical research base and is effective in managing acute suicidality. More so, in individuals with depressive disorders and suicidal behavior. [\[87\]](#) They have also been found to be effective in conditions where delaying treatment can be life-threatening, such as when a patient refuses to eat due to psychotic symptoms or in catatonia. Literature shows that ECT requires a shorter time to resolve the acute suicidality (vs antidepressants) in individuals with depression and suicidality as

antidepressants are known to take time to work.[55]

Moreover, it is the treatment of choice for pregnant females and where an individual is not tolerating the medications or not responding to initial lines of pharmacological or psychotherapy. Since there is a lack of evidence for the long-term benefit of ECT following acute phase treatment, other measures should be utilized in preventing chronic suicidality.

EFFECTIVENESS OF PSYCHOTHERAPY FOR SUICIDAL BEHAVIOR IN DEPRESSION

The role of the psychological intervention depends upon the specific aspects of the suicidal behavior (suicidal ideation, suicide attempt, completed suicide) for which a given intervention is effective. Suicidal ideations refer to thinking about, seriously considering, or planning suicide, while a suicide attempt represents a nonfatal self-directed, potentially injurious behavior, with an intent to end one's life, even if the behavior does not serve the intended purpose.[88] Psychotherapy is the cornerstone of managing suicidal behavior in depressed individuals. Various theories have been proposed that could predict the conversion of suicidal ideations into suicidal attempts (ideation-to-action theory), such as interpersonal theory, the integrated motivational-volitional model, and the three-step theory.[88]

Generic psychotherapy for depression has limited efficacy in improving co-occurring suicidal behavior.[25,89] Therefore, the psychotherapies specially developed for suicidal behavior, like CBT for suicide prevention (CBT-SP), or Dialectic Behavior Therapy (DBT) (particularly with comorbid personality disorders), or problem-solving therapy, must be considered.[84] In contrast, the effect of the psychotherapies for hopelessness in depression is larger and significant [Table 14]. Below, we have provided recommendations on various psychotherapies for the prevention of SIs and/or suicide attempts (SAs).

A seminal SR on observational studies (n = 13) by Mendez-Bustos *et al.* [84] found that CBT (N = 4) and DBT (N = 7) are most effective in preventing suicidal ideations in depressed individuals or individuals with personality disorders or SUDs. Another SR-MA has found that the highest evidence for treating suicidality in depression is available for

CBT, MB-CT, and Rational Emotive Behavior al Therapy (REBT), with greater focus on behavior al elements.[89] Other therapies that have been studied in a controlled trial format include psychodynamic, existential, and humanistic therapy, as well as life review, although with less robust evidence (also refer to [Table 13](#)).

PSYCHOTHERAPIES FOR SIS AND LEVEL OF RECOMMENDATION

CBT

CBT has level 1 evidence for the reduction of SIs in depressed individuals. For instance, an umbrella review (involving systematic analyses of the SR-MA) has found that CBT, by addressing maladaptive cognitive appraisals, negative emotions, and problematic behavior, brings about improvement in both suicide ideations and suicide attempts.[90]

Notably, CBT (involving weekly 16–20 sessions, each for about 45 minutes) has been found to improve suicidal behavior independent of their role for depression in individuals with depression and suicidality.[91] Moreover, suicidal ideations at the previous session have significant temporal predictive value for current depression, but not vice versa.

Likewise, a quasi-experimental study on the effectiveness of Internet-based CBT (Watts *et al.*, 2012)[92] for reducing both SIs and depressive symptoms found that online CBT is cost-effective and improves accessibility.

Given the current level of evidence for CBT in reducing suicide ideation in depression, it is *‘recommended for use’* for depression with SIs.

DBT

DBT, which is a well-established treatment for suicidal behavior in borderline personality disorders, has been adapted for suicidal behavior in depression and shown positive results. Typically, the DBT involves emotional regulation, distress tolerance, interpersonal effectiveness, and problem-solving, which is often coupled with phone coaching and ensuring follow-up visits.

For individuals with depression and suicidal ideations, both individual and group-based DBT have shown effectiveness in reducing suicidal ideations, which lasted for 12 months. An Irish program, with a quasi-experimental design, involving group and individual DBT, with weekly sessions, phone coaching, and follow-up visits with the treating consultant, has shown positive results in terms of reducing SIs.[93]

IPT

The role of the IPT in depression with suicidal behavior has been supported by controlled trials. A study by van Benthum *et al.* (2021)[91] has shown that IPT by intervening at the interface of the ideation-to-action framework (thwarted belongingness, perceived burdensomeness, hopelessness, suicidal desire, capability for suicide, suicide attempt) brings about a significant reduction in suicidal behavior (SIs and SAs) in depressed individuals, an effect independent of depression. Despite level 1 evidence, however, due to a limited number of well-controlled trials or lack of robust meta-analytic support, it is *conditionally recommended for use* (rather than *recommended for use*) with antidepressants for suicidal ideations in depression.

MB-CT

MB-CT has been found to be effective in treating depression and suicidal ideation in depression, albeit with less robust evidence. It works by equipping individuals to have more decentralized relationships with their negative thoughts, emotions, and sensations, thus preventing the downward spirals, where a negative affective state otherwise results in suicidal crises.[94]

MB-CT has been found to improve depression and hopelessness scores in depressed individuals with suicidality; however, its efficacy for improvement in suicidal ideations is not clinically significant. This underscores the need for more extensive MB-CT in depressed clients with suicidality, including addressing suicidal imagery more robustly.[94] While a few studies have found improvement in suicidal ideations in depressed individuals, for instance, Forkmann *et al.* (2014)[95] have found in their controlled trial, a significant reduction in suicidal ideations (SIs) in the MB-CT group versus the wait-list

control group. This change is found to be independent of the change in depression, rumination, and mindfulness skills. Another study from the same authors has found superiority of the MB-CT and Cognitive Behavioral Analysis System of Psychotherapy added to the treatment as usual (antidepressants) over the TAU alone in reducing suicidal ideations in depressed individuals, controlling for the improvement in depression symptoms.[96]

Family therapy

There are a few SR-MA that have shown the effectiveness of the family-focused therapy (e.g., attachment-focused family therapy, integrating the components of family therapy into CBT and DBT) over waitlist or an active control in reducing suicidal ideations and suicide attempts among the study participants.[97,98]

Family therapy in suicidal behavior targets reframing the relationship of patients with significant others, identifying and addressing the core conflicts, focusing on parental expression of love and empathy toward their wards, reattachment tasks wherein the patients work through emotions of shame, guilt, self-criticism, poor communication with family members, and gaining autonomy and taking up and exercising developmental responsibilities.[97]

However, this improvement has not reached a clinical significance level when it comes to improvement in depression. Moreover, the available data are largely limited to adolescents and the young adult population or specific vulnerable populations.[99] Hence, more research is required involving the middle-to-late adult population; more so, in the population experiencing suicidal behavior in the background of depression.

Literature has also shown that group therapy and Internet-based CBT therapy (supported by controlled design as well as quasi-experimental studies), along with ongoing psychosocial support measures, have proven to be cost-effective methods of SIs in individuals with depression.[100,101]

It must be highlighted that suicide prevention in depressed or borderline disorder patients

often involves a multimodal approach, involving intensive sessions, remote (e.g., telephone-based) support, and facilitating follow-up visits or treatment adherence. [102,103] The focus of therapy is often based on establishing a strong therapeutic alliance, collaborative assessment, ensuring patient engagement and adherence to treatment, wellness-promoting activities at home, and ensuring community support. [84]

Notably, apart from the structured and extensive psychotherapy for addressing suicidal behaviors (SIs and SAs) in the patients, available guidelines support intensive nonspecific OPD-based interventions as the mainstay of treatment, particularly when the availability of trained psychotherapists is limited. [84] However, in the available literature, including Systematic Reviews and Meta-Analyses on the topic, the type and extent of psychotherapies used have been heterogeneous, with the duration of psychotherapy ranging from a few weeks to months and the frequency of sessions varying from weekly to 1–2 sessions per week.

Problem-solving therapy

It has level 2 evidence for the prevention of suicidal ideation in depressed individuals. Likewise, an RCT has found significant effectiveness of the PST for reduction in suicide potential, including suicidal ideations and improvement in depression scores among the depressed adolescents and young adults as compared to a wait-list control. [104] A quasi-experimental study found that assertive brief psychotherapy (assertive patient-centered therapeutic engagement, crisis intervention, problem-solving strategies, and a targeting on community resources) delivered by the social worker and community linkage resulted in improvement in suicidal ideation at 6-month follow-up. It also resulted in a quality of life and depression. [100]

PSYCHOTHERAPIES FOR SUICIDAL ATTEMPTS (SAS) AND LEVEL OF RECOMMENDATION

Concerning the effectiveness of psychotherapy for suicide attempts (reduction in reattempt in an individual or the number of attempts over a period of time), CBT and DBT, with problem solving, hope induction, and coping skill training, seem to be

promising options for interventions with suicide attempts. However, heterogeneity of the studies in terms of the samples, treatment (types and frequency), comparator (waitlist, TAU, active comparator), outcome measures (self-report vs objective or hospital record-based), and sustenance of the effectiveness raises some concerns on the robustness of the findings.

The DBT has level 1 evidence (support of well-controlled RCTs) for suicide attempt prevention in depression. The key elements of the DBT to address suicidal attempts in depression include individual psychotherapy, group skills training, telephone coaching, and a consultation team for therapists.[103,105,106] DBT is particularly helpful for depressed individuals who have frequent suicidal tendencies and prominent anxiety. The efficacy of the DBT in depression with suicidal attempt has been supported by SR-MA as well.[84,107] It is particularly helpful with depressed individuals with prominent anxiety and frequent suicidal behaviors. However, the improvement is significant for a medium period (up to 6 months), but not sustained, lasting only 12 or 24 months. Although DBT has level 1 evidence for preventing suicide attempts in depressed individuals, the literature support is greater for borderline personality disorders (BPD) or depression in the context of BPD.[93] It has been found to be effective in both individual and group formats (indicating the cost-effectiveness in treatment).[84,108,109]

CBT for suicide prevention (CBT-SP)

CBT-SP, CBT with problem-solving and coping skill training, along with some component of emotional regulation, has level 1 evidence (well-controlled RCTs as well as meta-analytical support) for the suicide attempt prevention in depression.[110] Typically, CBT for suicidal patients targets maladaptive schemas and addresses issues like hopelessness, impulsivity, cognitive distortions, and problem-solving difficulties, thus helping individuals overcome all or none thinking. CBT has also been the core component of multimodal interventions for suicide prevention (particularly for individuals visiting the ERs with a crisis), such as Collaborative Assessment and Management of Suicide (CAMS). The latter involves a collaborative approach to suicide assessment and intervention, focuses on therapeutic alliance, cognitive restructuring, and self-monitoring/healthy thinking class.[103] Similarly, CBT forms an essential component of the Systems Training for Emotional Predictability and Problem-Solving (STEPPS);[111,112] here, CBT is combined with the

social skill training and systems components (continuous support by the reinforcement team or system members) to bring about an improvement in suicide reattempts (fewer suicide attempts at 6 months), better emotional regulation, and fewer hospitalizations. [\[112\]](#)

Problem-solving and contact

Problem-solving therapy (PST) or a problem-based approach to suicide prevention in depression has shown positive results, with a number of controlled and quasi-experimental (e.g., uncontrolled studies) establishing its effectiveness. PST equips individuals to comprehensively identify problems, generate potential solutions, perform cost-benefit analysis, and execute action plans. It improves coping, decreases impulsivity, and enhances hopefulness and self-efficacy, which are protective against suicide. It can be delivered in-person, individually or in a group, as well as telephonically or computer-based. [\[102,104\]](#)

For instance, a study by Alonzo *et al.*, 2016 [\[102\]](#) showed the effectiveness of the problem-solving approach to suicide prevention (manualized problem solving and comprehensive contact intervention – PS-CCI) among individuals with mood disorders visiting the emergency room (ER). This intervention includes problem-solving interviews at the ER (anticipated barriers to treatment identified and addressed, decisional balance worksheets), a suicide contract, and ensuring outpatient visits through reminders (postcard and telephonic contact). The intervention was found to be feasible, with good acceptability among the patients, greater retention in the treatment, and significantly less indulgence in suicide attempts.

Similarly, problem-solving-based interventions, such as OPAC (outreach, problem-solving, hope induction, adherence, and continuity in the phase acute, and motivation for further therapy), have shown a significant reduction in suicide reattempt at 12 months and reduced overall number of suicides attempts up to 60 months compared to the TAU group. [\[113,114\]](#)

Psychodynamic psychotherapy

The effectiveness of psychodynamic psychotherapy in suicide reattempt prevention has been supported by RCTs.[115] Typically, psychodynamic therapies target conscious interpersonal conflicts, traumatic experiences, past or continuing painful experiences, and self-harm attempts as a means to overcome/or get rid of these experiences.

For instance, a study by Guthrie *et al.* 2001[116] involving brief PDP suggests that psychoanalytic and psychodynamic psychotherapy significantly reduces (vs the control group) the suicide attempts in individuals with depressive disorders. However, this benefit persisted only up to 6 months. Additional benefits of such interventions include psychological wellbeing and fewer hospital admissions.

However, the research is often limited to borderline PD, with only a limited number of available trials, and the moderate quality of these trials warrants further research in this area.[115,117] Hence, in this guideline, we have kept it as conditionally recommended treatments for suicide attempt prevention in depressed individuals.

NONINVASIVE BRAIN STIMULATION IN DEPRESSION

Despite adequate trials of antidepressant treatment, psychotherapy, and various augmentation strategies involving medications, several persons with depression are still unable to attain remission. In such scenarios, noninvasive brain stimulation techniques like ECT, repetitive transcranial magnetic stimulation (rTMS), and transcranial direct current stimulation (tDCS) are increasingly used in clinical practice.

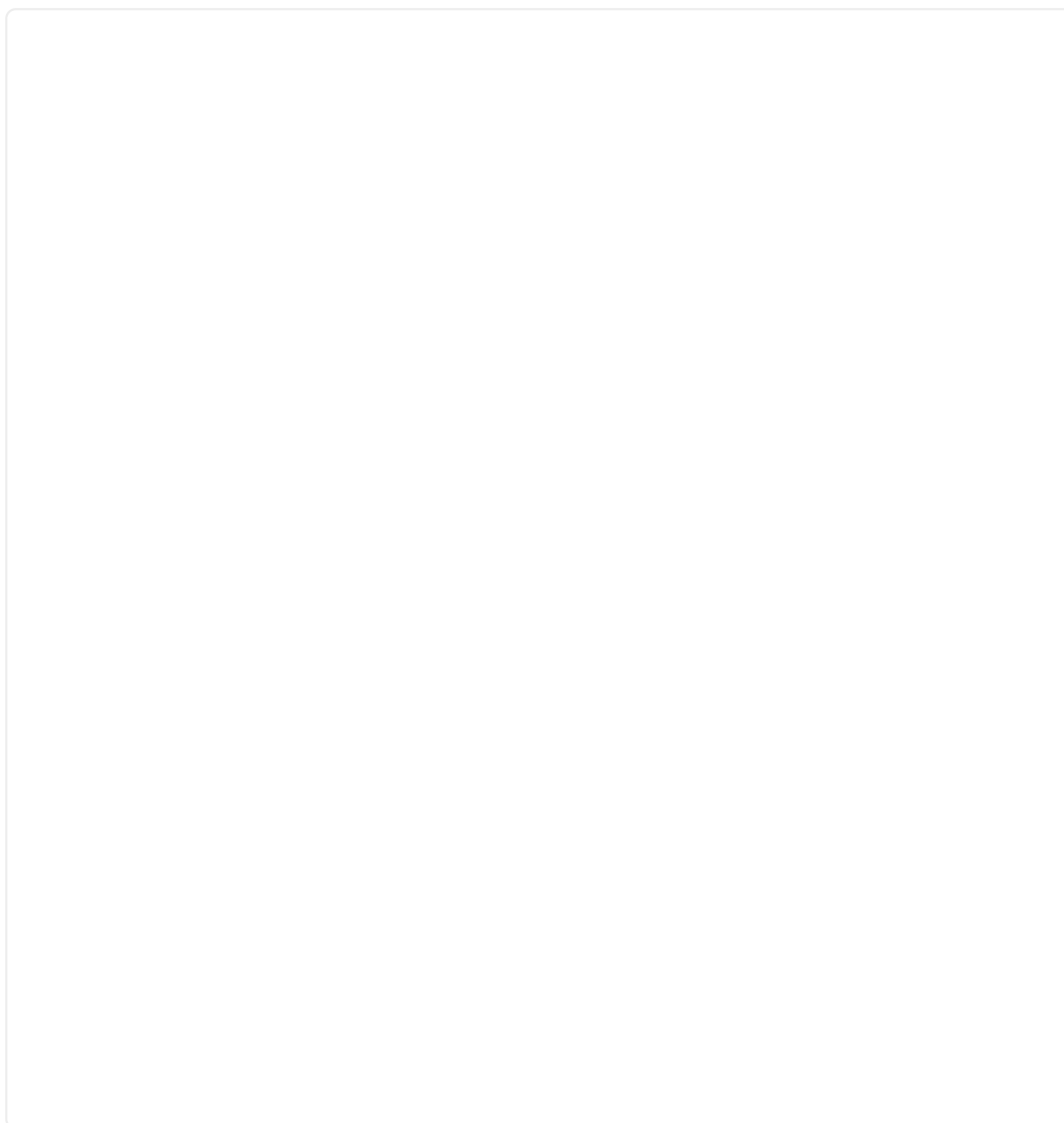
[118,119,120,121,122] In most conditions, these neuromodulation techniques are used as add-on strategies to the ongoing pharmacological and/or psychological treatments for depression.

ELECTROCONVULSIVE THERAPY FOR DEPRESSION

ECT has been extensively used in the management of depression. Treatment-resistant depression is one of the common indications of ECT.[118,121] ECT is often considered as a second- or third-line treatment option for depression.[121] However, in emergency scenarios where there is suicidal behavior associated with a depressive episode, coexisting

psychotic symptoms, poor oral intake, catatonia, and severe physical debilitation associated with depression, ECT is considered as a primary treatment modality.[\[118,121\]](#) Similarly, depression associated with the peripartum period, significant distress warranting rapid symptom resolution, and those with severe mixed affective episodes are indications of ECT.[\[121\]](#) The figure below summarizes the predictive factors associated with better response to ECT [\[Figure 2\]](#).

Figure 2.



[Open in a new tab](#)

Factors associated with better response to ECT in depression

Nearly half of the patients with depression relapse after achieving remission when ECT is abruptly discontinued.[118] This warrants the need for maintenance ECT to prevent relapse of the symptoms. Another major drawback of ECT is the lack of a specific

focus/target area.[118] To overcome this, more targeted interventions like TMS and tDCS are gaining the attention of clinicians and researchers.

TRANSCRANIAL MAGNETIC STIMULATION FOR DEPRESSION

TMS is an effective modality of treatment for depression, which is more of a focal intervention and uses a magnetic stimulus generated from an electric current while traversing through the coil. The magnetic stimulus gets converted to electrical stimulus again on reaching the cortical surface (due to the principle of electro-magnetic induction). [118] In 2008, the US FDA had approved high-frequency rTMS targeting the left dorsolateral prefrontal cortex (left DLPFC) at 120% of resting motor threshold for the treatment of depression,[118,122,123,124] which is completed in over 38 minutes for a single session. With technology advancement, theta burst stimulation has been developed, and the theta burst protocol delivers the session in just a little more than 3 minutes with equivalent safety and efficacy.[118] This protocol has recently been approved by the US FDA. High frequency (more than 5 Hertz) rTMS and intermittent theta burst stimulation produce an excitatory effect on the underlying cortex, whereas low frequency (1 Hertz and less) rTMS produces an inhibitory effect on the underlying cortex.[122,123,125] [Figure 3](#) gives a flow of steps in delivering TMS in depression.

Figure 3.

[Open in a new tab](#)

Steps to deliver TMS in depression

Several TMS protocols have been approved by the US FDA for use in depressive disorder based on available evidence. [Table 17](#) below summarizes the US FDA-approved protocols of TMS in the management of depression.

Table 17.

US FDA-approved TMS protocols for depression

Year of FDA approval	Target area	Protocol	Number of
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			sessions
2008[126 , 127 , 128]	Left dorsolateral prefrontal cortex	10 Hz frequency, 40 pulses/train, 26 seconds intertrain interval, 75 trains, 3000 pulses/session, 37.5 minutes, 120% of resting motor threshold	5 sessions/week for 4–6 weeks (20–30 sessions)
2013[127 , 128 , 129]	Deeper brain areas (bilateral dorsolateral prefrontal cortex)	Deep TMS using H1 coil	5 sessions/week for 4–6 weeks (20–30 sessions)
2018[126 , 127 , 128]	Left dorsolateral prefrontal cortex	Intermittent theta burst stimulation (600 pulses/session over a period of 3 minutes and 9 seconds)	5 sessions/week for 4–6 weeks (20–30 sessions)
2022[126 , 127 , 130 , 131]	Left dorsolateral prefrontal cortex	Stanford Accelerated Intelligent Neuromodulation Therapy (SAINT) protocol: 1800 pulses/session, 50-minute intersession interval, intermittent theta burst stimulation at 90% of resting motor threshold	10 sessions/day and 50 sessions over 5 consecutive days

[Open in a new tab](#)

Other than this, there are several other non-US FDA-approved protocols used in the treatment of depression. Low-frequency rTMS (1 Hz) and continuous theta burst stimulation have been used targeting the right dorsolateral prefrontal cortex in the management of depression.[\[126\]](#) The common side effects are headache and numbness of the head, which are mostly self-limiting. TMS needs to be avoided in patients with intracranial ferromagnetic implants (e.g., cochlear implant, aneurysmal clips, electrodes for deep brain stimulation).[\[125\]](#) Seizure is the most serious adverse effect associated with TMS intervention; however, it is rare.

TRANSCRANIAL DIRECT CURRENT STIMULATION FOR DEPRESSION

Transcranial direct current stimulation (tDCS) uses low-intensity electric current for modulating cortical activity. Anodal stimulation produces underlying cortical excitation by depolarizing the neurons, whereas cathodal stimulation produces underlying cortical inhibition by hyperpolarizing the neurons. Repeated application of tDCS produces a sustained antidepressant effect, resulting from neuroplastic changes in specific brain neurocircuits.[\[120\]](#) Several other transcranial electrical stimulation modalities [transcranial alternating current stimulation (tACS), transcranial random noise stimulation (tRNS), and transcranial pulsed current stimulation (tPCS)] are used in the management of neuropsychiatric disorders, including depression. However, the evidence regarding the use of tACS, tRNS, and tPCS is preliminary.

The common target of tDCS in the treatment of depression is the dorsolateral prefrontal cortex (dlPFC).[\[120,132,133\]](#) tDCS is delivered by anodal placement over the left dlPFC and cathodal placement over the right dlPFC with the use of electrodes with surface areas of 25 to 35 cm² and 2 mA electric current strength over 20 to 30 minutes.[\[120,133\]](#)

tDCS is well tolerated, and adverse effects (headache, numbness, burning sensation at the site of stimulation) are transient and mostly self-limiting.[\[133,134,135\]](#) Recently, home-based treatments, which are remotely operated by the clinician and self-administered treatment of tDCS, are gaining public attention.[\[136,137\]](#)

LEVELS OF EVIDENCE OF NEUROMODULATION IN DEPRESSION

TREATMENT AND PRACTICE RECOMMENDATIONS

As per the available evidence, the short-term efficacy, safety, and tolerability of rTMS and ECT are well established; however, for maintenance treatment, ECT has the highest level of evidence.[124] The current recommendation regarding the use of TMS in the management of depression is a failed trial of one or more antidepressants and psychotherapy.[123] A recent network meta-analysis revealed that bitemporal ECT, high-dose right unilateral ECT, transcranial magnetic stimulation with priming, magnetic seizure therapy, bilateral rTMS, bilateral theta burst stimulation, low-frequency right rTMS, intermittent theta burst stimulation, high-frequency left rTMS, and tDCS all demonstrate greater efficacy compared to sham intervention.[132] These modalities of neuromodulation may be considered as an add-on treatment in depression. Efficacy-wise, ECT is superior to TMS and tDCS, whereas in terms of safety, tDCS and TMS are safer than ECT.[132] The clinician needs to consider the following practice recommendations while using neuromodulation treatment in depression:

1. Informed consent from the patient (if possible, from the caregivers, as well) to enroll the patient for neuromodulation treatment.
2. Evaluating and adjusting the dose of the medications that alter cortical excitability
3. Evaluate the indications and contraindications of neuromodulation in the patient.
4. Selection of an appropriate protocol.
5. Appropriate selection of the cortical target area for the given psychiatric condition.
6. Identification of the target area of neuromodulation by using a standard technique.
7. Periodic monitoring of adverse effects and therapeutic effects.

It is important to understand when to continue the neuromodulation treatment and when to stop it, when it is given to a patient with depression. [Table 18](#) below summarizes the conditions under which neuromodulation sessions should be continued and the conditions under which they should be stopped.

Table 18.

Recommendations for continuing and stopping the neuromodulation sessions in depression

Recommendations for continuing the neuromodulation sessions in depression	Recommendations for stopping the neuromodulation sessions in depression
• Continuing clinical benefits (objective improvement in the rating scale scores) • Relapse of symptoms on discontinuation of neuromodulation	• Minimal and no response to neuromodulation • Serious/intolerable adverse effects encountered with neuromodulation • Worsening of the symptoms of depression following neuromodulation sessions • Patient is not willing to continue neuromodulation treatment further (withdrawal of consent)

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CONCLUSION

The 2025 update of the depression guidelines offers an updated, culturally informed, contextually relevant, and personalized framework for the management of depressive disorders among adults in India. These recommendations are designed to assist psychiatrists in providing person-centered, evidence-based care across all stages of the illness from early identification and assessment to acute treatment, continuation, and long-term maintenance. By integrating global evidence with indigenous clinical experience, the updated guidelines aim to enhance functional recovery, prevent relapse,

and improve overall quality of life for individuals with depression.

Conflicts of interest

There are no conflicts of interest.

Acknowledgement

The authors would like to express their sincere gratitude to Ms. Pritanshi Jeswani, MPhil (Clinical Psychology), for her valuable support in proofreading, reviewing, and assisting with the final draft of this manuscript.

Funding Statement

Nil.

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